STOCKSENSE

ENGINEERING KPI TARGET SHEET

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> **Verification Overview:** This KPI Target Sheet defines explicit quantitative baselines, acceptable operational thresholds, and standardized test verification methods for StockSense across hardware, firmware, power, networking, and predictive analytics layers.

1. HARDWARE & SENSING ARCHITECTURE KPIS

METRIC ID	ENGINEERING PARAMETER	BASELINE TARGET	MAX THRESHOLD	VERIFICATION TEST METHODOLOGY
KPI-HW-01	Load Cell Measurement Accuracy	**≤ ±1.0% FS**	≤ ±2.0% FS	Multi-point calibration with standard certified weights (50g–10kg) across 20 test runs.
KPI-HW-02	ADC Dynamic Resolution	**24-Bit Diff**	24-Bit Raw	Noise-floor FFT analysis on differential bridge voltage at zero-load condition.
KPI-HW-03	Weight Thermal/Time Drift	**< 0.5% / 24h**	< 1.0% / 24h	Thermal chamber soak test (10°C to 45°C) over 24-hour cycle with static load.
KPI-HW-04	Signal Settling Latency	**≤ 500 ms**	≤ 1000 ms	Time-domain response measurement from weight contact to filter steady state.
KPI-HW-05	Prototype Node BOM Cost	**≤ LKR 10,000**	≤ LKR 12,000	Component procurement cost audit across PCB, sensors, power, and enclosure.

2. FIRMWARE & EDGE PROCESSING KPIS

METRIC ID	ENGINEERING PARAMETER	BASELINE TARGET	MAX THRESHOLD	VERIFICATION TEST METHODOLOGY
KPI-FW-01	Active Sensing Sampling Rate	**10 Hz**	5 Hz – 10 Hz	MCU timer interrupt loop benchmark during weight event transitions.
KPI-FW-02	Local Anomaly Response Time	**≤ 100 ms**	≤ 250 ms	Oscilloscope trace from weight drop to local alert output pin activation.
KPI-FW-03	Offline Flash Buffer Capacity	**≥ 10,000 Recs**	≥ 5,000 Recs	Simulated Wi-Fi disconnect test writing log events to local SPI Flash.
KPI-FW-04	Re-sync Dispatch Rate	**≥ 50 pkts/s**	≥ 20 pkts/s	Throughput measurement when flushing buffered logs to MQTT upon Wi-Fi recovery.
KPI-FW-05	Firmware Memory Footprint	**≤ 70% Max**	≤ 85% Max	Static and dynamic heap/stack static allocation build toolchain audit.

3. SYSTEM TARGET SUMMARY HIGHLIGHTS

≤ ±1.0%	≤ 2.0s	≥ 8.0 Hrs	< 15%
SCALE PRECISION	MQTT TELEMETRY	BATTERY BACKUP	FORECAST ERROR

4. POWER SUBSYSTEM & RELIABILITY KPIS

METRIC ID	ENGINEERING PARAMETER	BASELINE TARGET	MAX THRESHOLD	VERIFICATION TEST METHODOLOGY
KPI-PWR-01	Battery Failover Runtime	**≥ 8.0 Hours**	≥ 6.0 Hours	Discharge test on internal Li-ion battery with continuous sampling and 1-min Wi-Fi ping.
KPI-PWR-02	Deep Sleep Current Draw	**≤ 10 mA**	≤ 25 mA	Precision current-shunt meter analysis during deep sleep power state.
KPI-PWR-03	Mains Outage Switchover	**≤ 10 ms**	≤ 50 ms	Power rail oscilloscope trace during 5V supply disconnect (zero reboot requirement).
KPI-PWR-04	Environmental Operating Range	**0°C–50°C**	5°C–45°C	Environmental chamber testing for thermal stability and high-humidity performance.

5. CLOUD CONNECTIVITY & ANALYTICS KPIS

METRIC ID	ENGINEERING PARAMETER	BASELINE TARGET	MAX THRESHOLD	VERIFICATION TEST METHODOLOGY
KPI-NET-01	MQTT End-to-End Latency	**≤ 2.0 Seconds**	≤ 5.0 Seconds	Round-trip timing from edge weight event to web UI dashboard render over Wi-Fi.
KPI-NET-02	Packet Delivery Success (QoS 1)	**≥ 99.5%**	≥ 98.0%	24-hour transmission audit comparing published messages vs. database received records.
KPI-AI-01	Run-Out Date Forecast Error	**< 15% Error**	< 20% Error	Back-testing ML models against historical consumption datasets over 30 simulated days.
KPI-UI-01	Dashboard Render Speed	**≤ 1.5 Seconds**	≤ 3.0 Seconds	Lighthouse browser performance audit loading multi-node live web UI.

Sign-off Note: These KPI engineering targets serve as the official benchmark against which Phase 2 (Breadboard Prototype) and Phase 3 (Custom PCB) bring-up testing will be evaluated during Gate Reviews.